



Matteo Merler

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Date of birth: 17/07/2000 **Nationality**: Italian

ABOUT ME

I am a pre-doctoral researcher at FBK NLP in Trento, Italy, with an MSc in Machine Learning, Data Science, and Artificial Intelligence from Aalto University in Helsinki. My research focuses on bridging large-scale models (such as LLMs and VLMs) with methods like Reinforcement Learning (RL) and planning. I aim to explore how RL's ability to learn from experience can complement the strengths of LLMs, which currently lack this capacity. I'm particularly interested in model-based RL and world models, investigating how large-scale models can be trained to plan ahead and imagine future states, helping them to adapt to open-ended reasoning and long-horizon planning beyond pattern matching.

WORK EXPERIENCE

Fondazione Bruno Kessler nlplab.fbk.eu

Address: Via Sommarive 18, 38123, Trento, Italy

[07/01/2025 – Current] **Researcher**

Working full-time as a pre-doctoral researcher in the NLP group under the supervision of Bernardo Magnini. My current research interests lie at the intersection of Reinforcement Learning, formal planning and Language/Vision Language Models.

- Lead researcher on FBK's side of a joint project on planning with Vision-Language Models (with Aalto University): the resulting work is available as a preprint: **ViPlan: A Benchmark for Visual Planning with Symbolic Predicates and Vision-Language Models** (<https://arxiv.org/pdf/2505.13180>).
- Led a second project on selectively integrating VLM guidance with model-free Reinforcement Learning based on the agent's own uncertainty, manuscript currently under review.
- Participating on private research for collaborations with companies.
- Contributing to the draft of a Horizon Europe funding proposal, leading my lab's team for Language and Reinforcement Learning.
- Active participation to the research group's scientific activities, including monthly meetings and collaborative discussions.

Aalto University

Address: Konemiehentie 2, 02150, Espoo, Finland

[01/08/2024 – 30/09/2024] **Research Assistant**

Full time research assistant position at the Foundation Models for Language and Reinforcement Learning (Larel) lab under prof. Pekka Marttinen for a research project advancing the work from my Master's Thesis. I was also working within the lab part-time for my Master's Thesis and for other research projects since September of 2023, but without formal employment status as I was still conducting my studies.

- Conducted a research project on Symbolic Regression with Large Language Models, presented at the ACL Student Research Workshop.
- Co-led a research project on Code World Models with PhD candidate Nicola Dainese, forming the basis of my Master's Thesis, accepted at NeurIPS 2024.

- Started a project on planning with Vision-Language Models, later continued by both Aalto University and my current group at FBK.
- Regularly attended lab meetings and participated in broader academic activities, including the Seminar on Large Language Models.

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Address: Via Antonio da Trento 8, 38122, Trento, Italy

[01/02/2022 – 30/06/2022] **Bachelor's Thesis Internship**

Formative internship in collaboration with the University of Trento. I analyzed, designed and built a prototype for an indoor navigation system specifically targeted for supermarkets, to be used with the company's own location tracking system:

- Analyzed the problem and available data coming from the company's deployments and tracking system.
- Implemented a dynamic Python prototype adapting state of the art routing systems to react in real time to store conditions.
- Gathered performance data and evaluated the results as a part of my Bachelor's Thesis work.

During my first real working experience I learned invaluable collaboration skills by working with great

teammates and mentors. Overseeing a project from the beginning to the end I was able to get familiar with

the proper steps of analyzing and prototyping and deliver a working program at the end of my tenure.

EDUCATION AND TRAINING

[01/09/2022 – 30/09/2024] **MSc in Machine Learning, Data Science and Artificial Intelligence**

Aalto University <https://www.aalto.fi/en>

Address: Konemiehentie 2, 02150, Espoo, Finland | **Final grade:** 5/5 GPA, degree obtained with honors. | **Type of credits:** ECTS | **Number of credits:** 120 | **Thesis:** Generating Code World Models with Large Language Models Guided by Monte Carlo Tree Search

During my studies, I focused on the use of Large Language Models (LLMs) with methods such as Reinforcement Learning (RL) or evolutionary-inspired search to create agents that can learn from experience and solve open-ended problems.

My Master's Thesis was **awarded as one of the three best Theses at the Aalto University School of Science in 2024.**

Publications:

- **In-Context Symbolic Regression: Leveraging Large Language Models for Function Discovery** (<https://arxiv.org/abs/2404.19094>), published in the ACL 2024 Student Research Workshop.
- **Generating Code World Models with Large Language Models Guided by Monte Carlo Tree Search** (<https://arxiv.org/abs/2405.15383>), published at NeurIPS 2024.

Relevant courses:

- CS-E4890 Deep Learning
- ELEC-E5550 Statistical Natural Language Processing
- ELEC-E8125 Reinforcement Learning
- CS-E4710 Machine Learning: Supervised Methods
- CS-E4820 Machine Learning: Advanced Probabilistic Methods D

[16/09/2019 – 21/07/2022] **BSc in Computer Science**

University of Trento <https://www.unitn.it/en>

Address: Via Sommarive 9, 38123, Trento, Italy | **Final grade:** 110 cum laude/110 |

Type of credits: ECTS | **Number of credits:** 180 | **Thesis:** Route Optimization for Indoor Spaces (in Italian)

Relevant courses:

- Introduction to Machine Learning
- Algorithms and Data Structures
- Geometry and Linear Algebra
- Probability and Statistics

[01/09/2014 – 04/07/2019]

High School Diploma

Liceo Scientifico Leonardo da Vinci <https://liceodavincitn.it>

Address: Via Cristoforo Madruzzo 24, 38122, Trento, Italy | **Final grade:** 100/100

HONOURS AND AWARDS

[05/03/2025]

Master's Thesis Award Awarding institution: Aalto University School of Science

Received the outstanding Master's Thesis Award from the School of Science. In 2024, 517 master's degrees were completed in the School of Science. In March, three Master's Theses out of these 517 were awarded.

Link: <https://www.aalto.fi/en/news/school-of-science-awards-outstanding-doctoral-and-masters-theses#1-master-s-thesis-awards>

PUBLICATIONS

[2025]

[ViPlan: A Benchmark for Visual Planning with Symbolic Predicates and Vision-Language Models](#)

We develop a benchmark for planning with Vision-Language Models (VLMs), including a setup where the VLM directly generates the plan and one where it is used to ground a formal symbolic planner.

Authors: Matteo Merler, Nicola Dainese, Minttu Alakuijala, Giovanni Bonetta, Pietro Ferrazzi, Yu Tian, Bernardo Magnini, Pekka Marttinen | **Journal Name:** ArXiv

[2024]

[In-Context Symbolic Regression: Leveraging Large Language Models for Function Discovery](#)

We propose a novel approach to Symbolic Regression that leverages Large Language Models (LLMs) to propose candidate functions that are iteratively scored and refined using the LLM itself.

Authors: Matteo Merler, Katsiaryna Haitsiukevich, Nicola Dainese, Pekka Marttinen | **Journal Name:** Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics | **Volume, Issue and Pages:** Volume 4: Student Research Workshop, pages 427–444

[2024]

[Generating Code World Models with Large Language Models Guided by Monte Carlo Tree Search](#)

We investigate the use of Large Language Models (LLMs) for generating world models for Reinforcement Learning. Our approach uses the LLM to generate a Python program that acts as a simulator for the environment, using a novel code generation strategy we call GIF-MCTS.

Authors: Nicola Dainese, Matteo Merler, Minttu Alakuijala, Pekka Marttinen | **Journal Name:** Advances in Neural Information Processing Systems 37 | **Volume, Issue and Pages:** Pages 60429–60474

VOLUNTEERING

[01/07/2017 – 31/07/2017]

Libera Associazioni Nomi e Numeri Contro le Mafie Reggio Calabria, Italy

I participated in a volunteering camp experience in southern Italy where we cultivated the fields of mafia confiscated house for the Libera association, while also taking part to various activities on the territory and listening to testimonies from victims and policemen.

Link: https://www.libera.it/it-schede-1326-libera_inglese

[01/08/2018 – 31/08/2018]

Associazione Volontaria "Per un Mondo Migliore" Rijeka, Croatia

I participated in a volunteering camp experience in Rijeka, Croatia where we helped different families and associations in need around the city and in mainland Croatia while living together in a shared home.

Link: <https://betterworldcroatia.org>

SKILLS

Scientific Computing

Python | Conda | PyTorch | Jupyter | Julia | R | Git | SLURM | LaTeX | Docker/ Apptainer

Foundation Model Training/Finetuning/Inference

Huggingface (Transformers, Accelerate, PEFT, TRL) | DeepSpeed | vLLM

LANGUAGE SKILLS

Mother tongue(s): Italian

Other language(s):

English

LISTENING C2 READING C2 WRITING C2

SPOKEN PRODUCTION C2 SPOKEN INTERACTION C2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user